

28. THE KNEELING BODY, LATERAL

DURATION : 17:52

STUDY SHEET

COURSE SUMMARY

Study of the role of the lateral geniculate body in processing visual information, via its parvocellular and magnocellular layers (processing of colour and movement). Analysis of synaptic transmission via the optic nerve to the cortical areas and interaction with the pulvinar. The presentation illustrates multisensory integration and its links with Papez's circuit, demonstrating how visual perception is modulated by memory patterns.

THE LATERAL GENICULATE BODY: INTEGRATION AND PROCESSING OF VISUAL INFORMATION

The **lateral geniculate body** plays a crucial role in processing visual information. It receives nerve impulses from the retina and sorts this information before transmitting it to the visual areas.

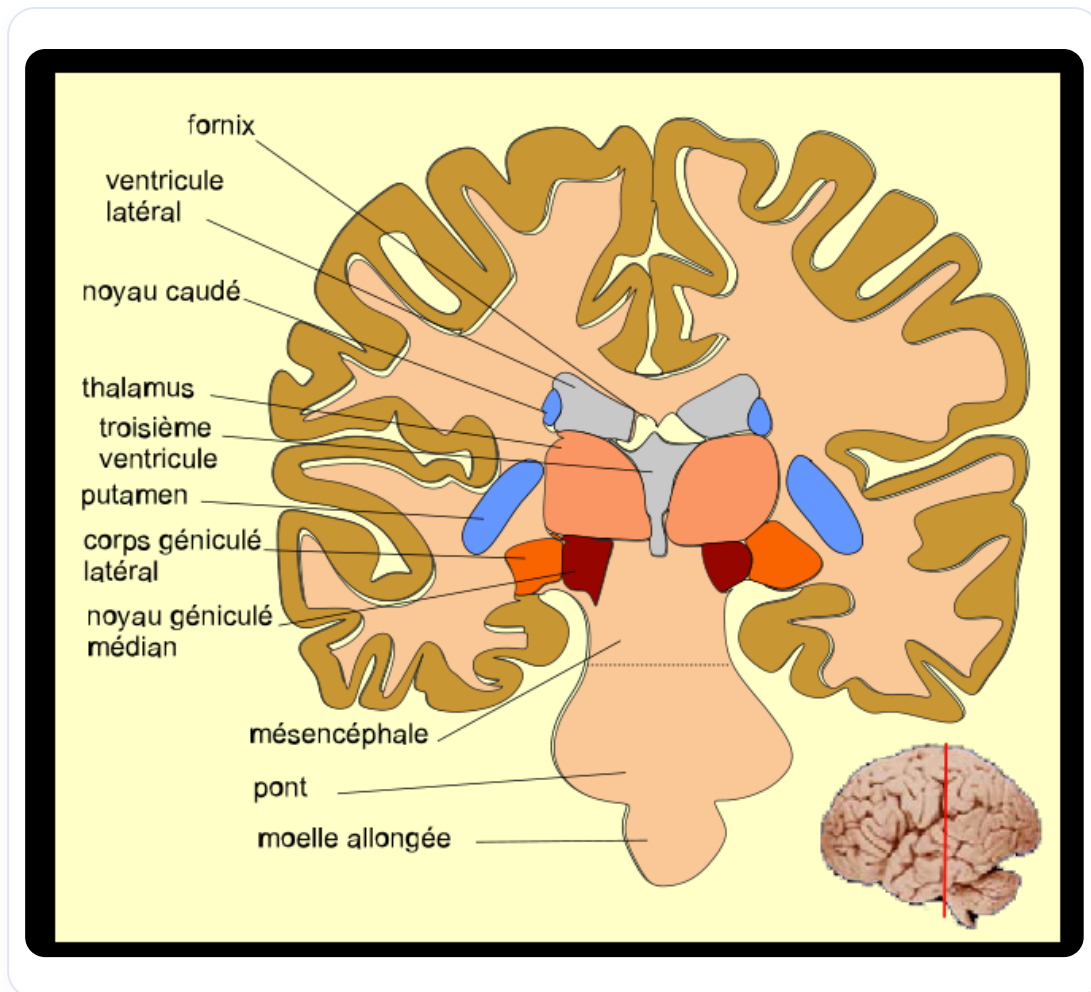


FIG. — Corps Genouillé Latéral (CGL)

STRUCTURE AND FUNCTION

The lateral geniculate body is composed of several layers, each specialised in processing specific information. There are mainly three types of cells:

- **Parvocellular cells:** These process information related to **colour** and **form**.
- **Magnocellular cells:** These are responsible for detecting **movement** and **orientation**.
- **Koniocellular cells:** These focus on **spatial frequencies**.

Each layer of the lateral geniculate body contributes to the formation of an image by integrating these different pieces of information.

INFORMATION TRANSMISSION

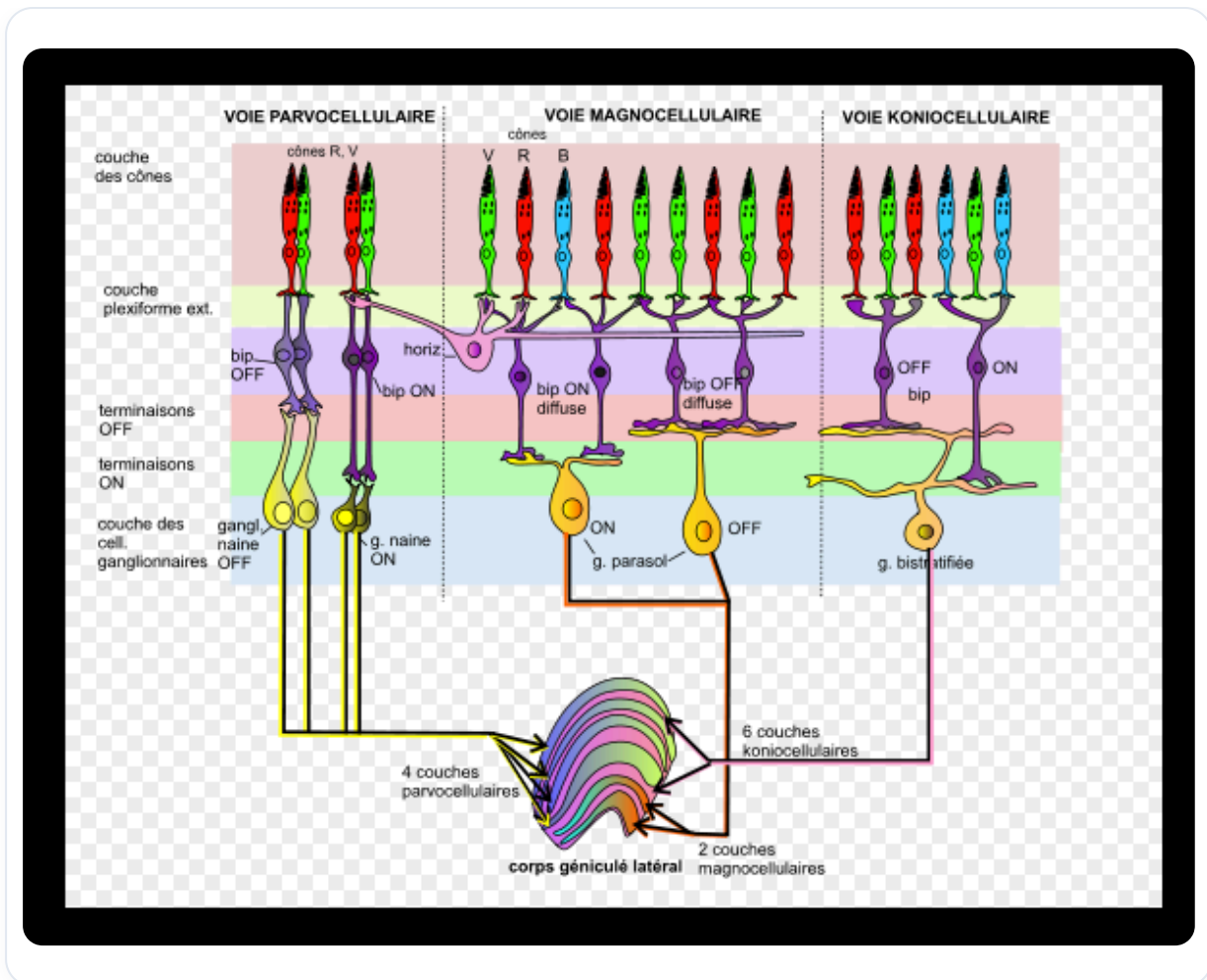


FIG. — Parvocellular and magnocellular cells

Visual information is transmitted through the optic nerve and arrives at the lateral geniculate body, where it is sorted before being sent to the thalamus and visual areas. Approximately **80%** of the information processed by the lateral geniculate body is directly sent to the visual areas, while **20%** passes through the pulvinar, another thalamic structure, for more subtle processing.

The pulvinar also receives information from other nuclei, such as the **superior colliculus** and the **olivary nuclei**, which are essential for integrating auditory and visual information.

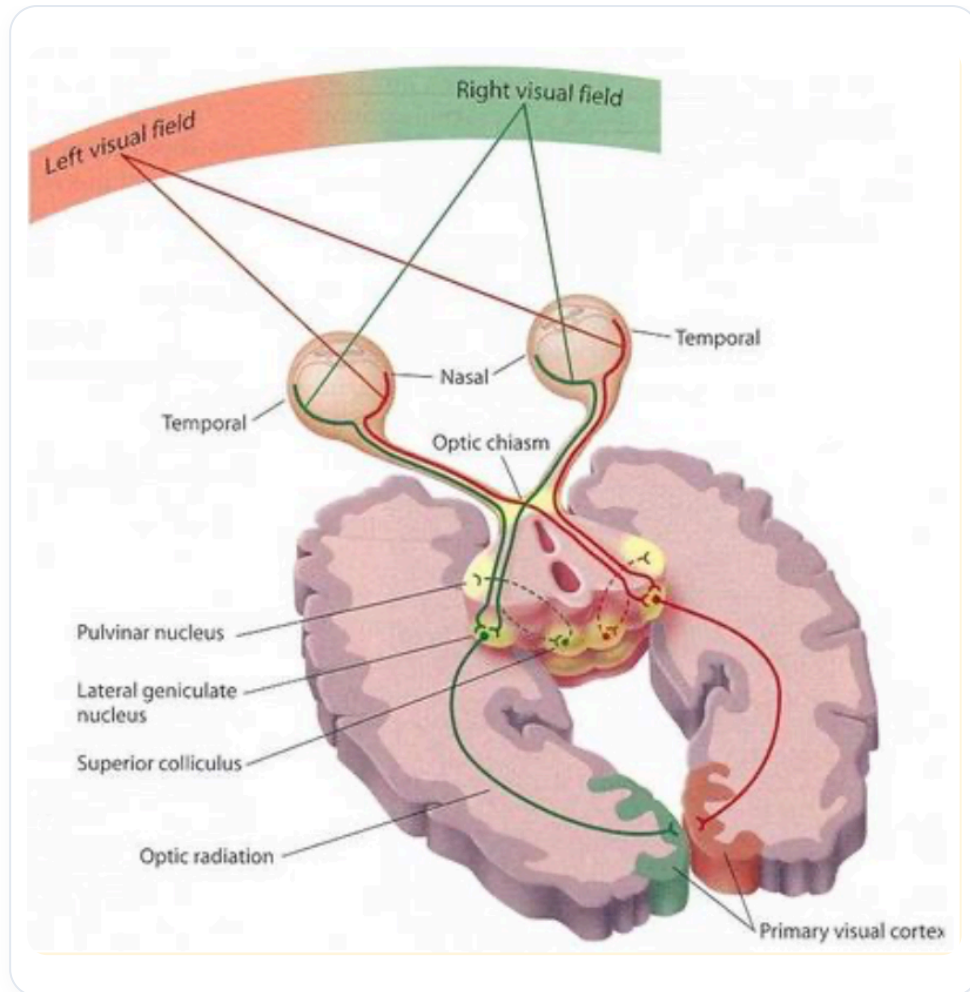


FIG. — Information distribution from LGN to pulvinar

MULTISENSORY INTEGRATION

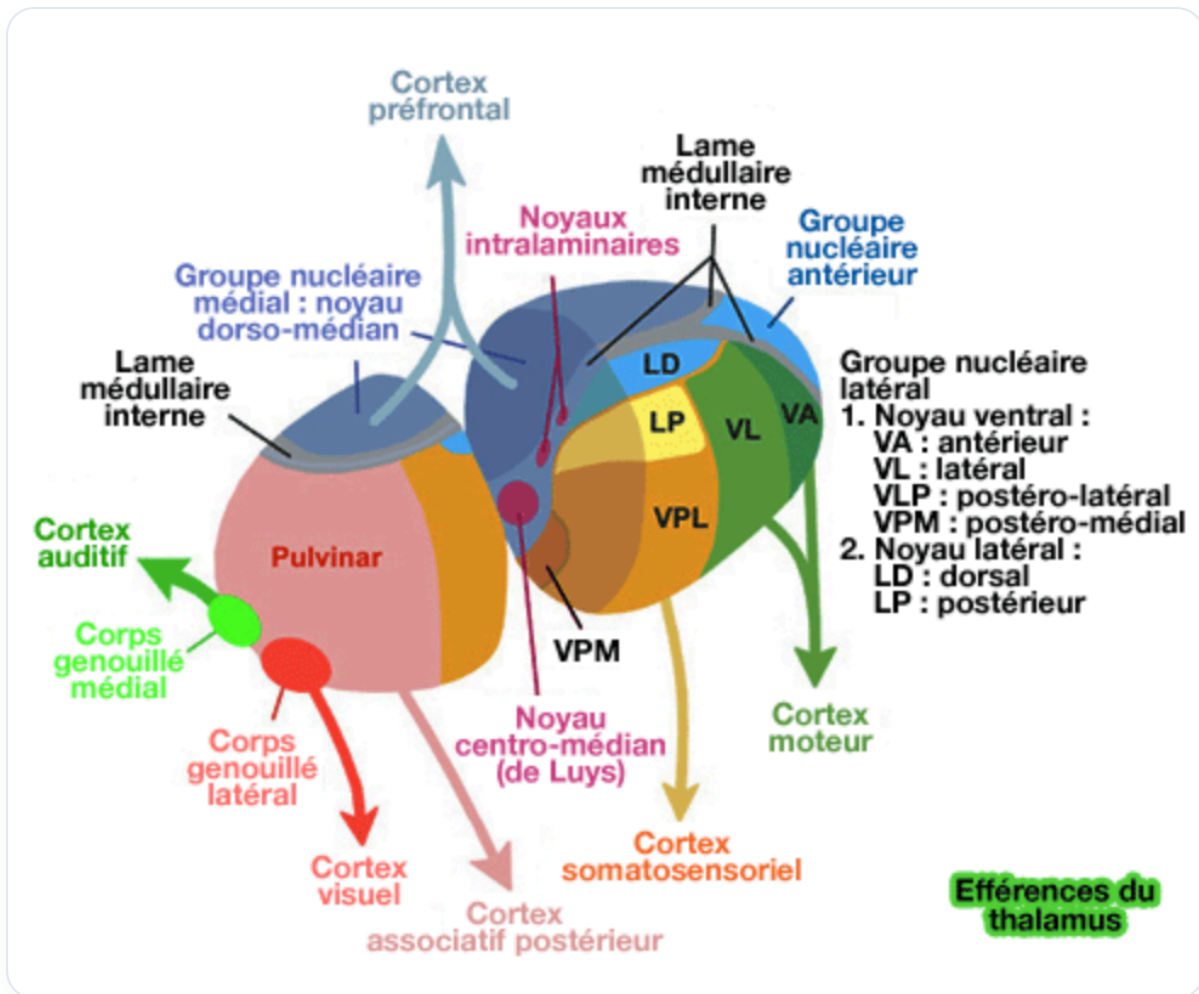


FIG. — Pulvinar connections

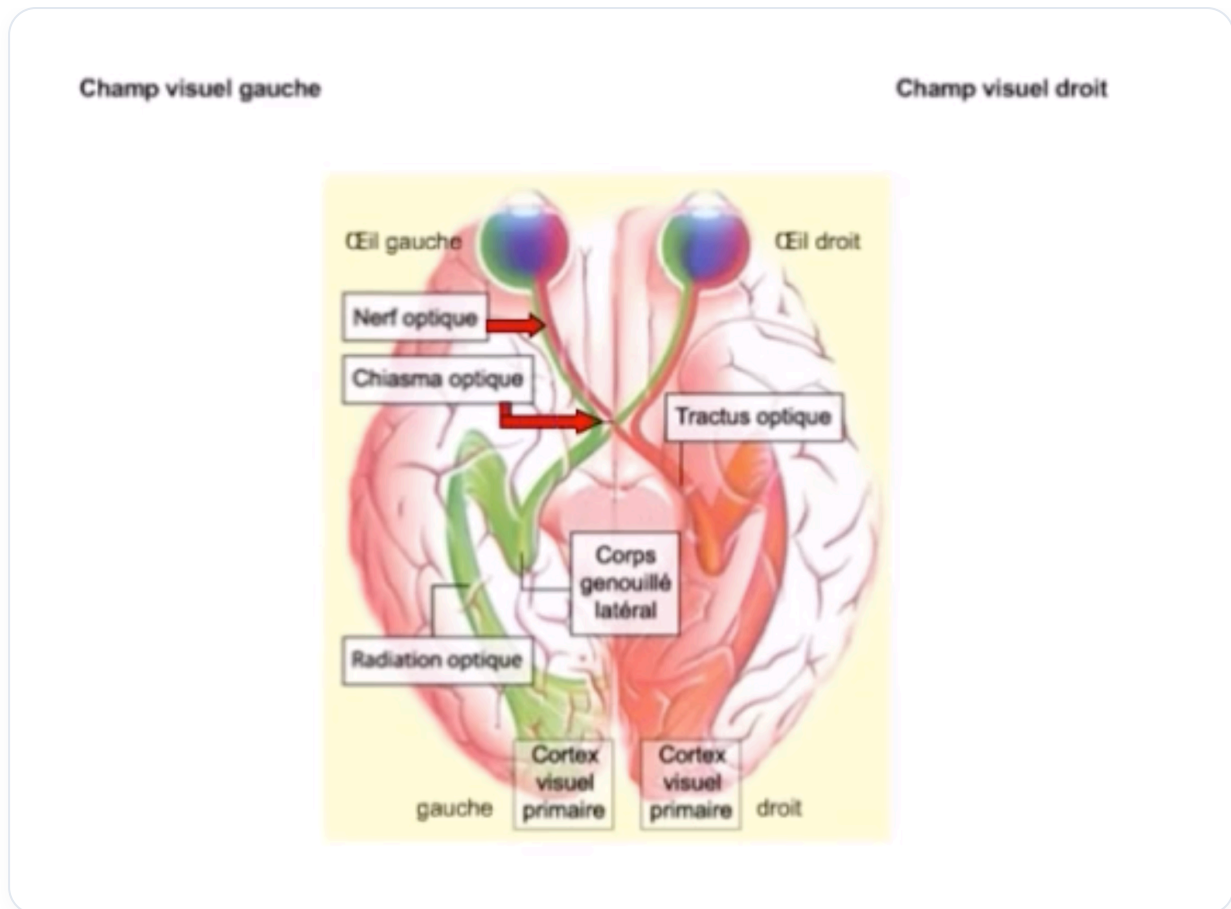


FIG. — Visual information pathway to the LGN

The lateral geniculate body not only processes visual information but is also connected to auditory and vestibular systems. This allows for the integration of information from the eyes, ears, and the body's position in space. This process is essential for actions such as ocular **saccades** and head orientation.

PAPEZ CIRCUIT AND MEMORY

- Le circuit de Papez est un ensemble de structures nerveuses du cerveau impliquées dans la mémorisation et la formation de la mémoire à long terme.
- Tractus hippocampo-mammillo-thalamo-cingulaire
- Bilatéral et symétrique

MATCH

M: Mammillary bodies

A: Anterior, **T:** Thalamus

C: Cingulate Gyrus

H: Hippocampus

FIG. — Intégration multisensorielle (auditive et vestibulaire)

The lateral geniculate body is also linked to the **Papez circuit**, which plays a role in memory. This circuit includes structures such as the **mammillary bodies**, the **anterior thalamus**, and the **cingulate cortex**. It is involved in the storage and retrieval of memories, linking visual information to past experiences.

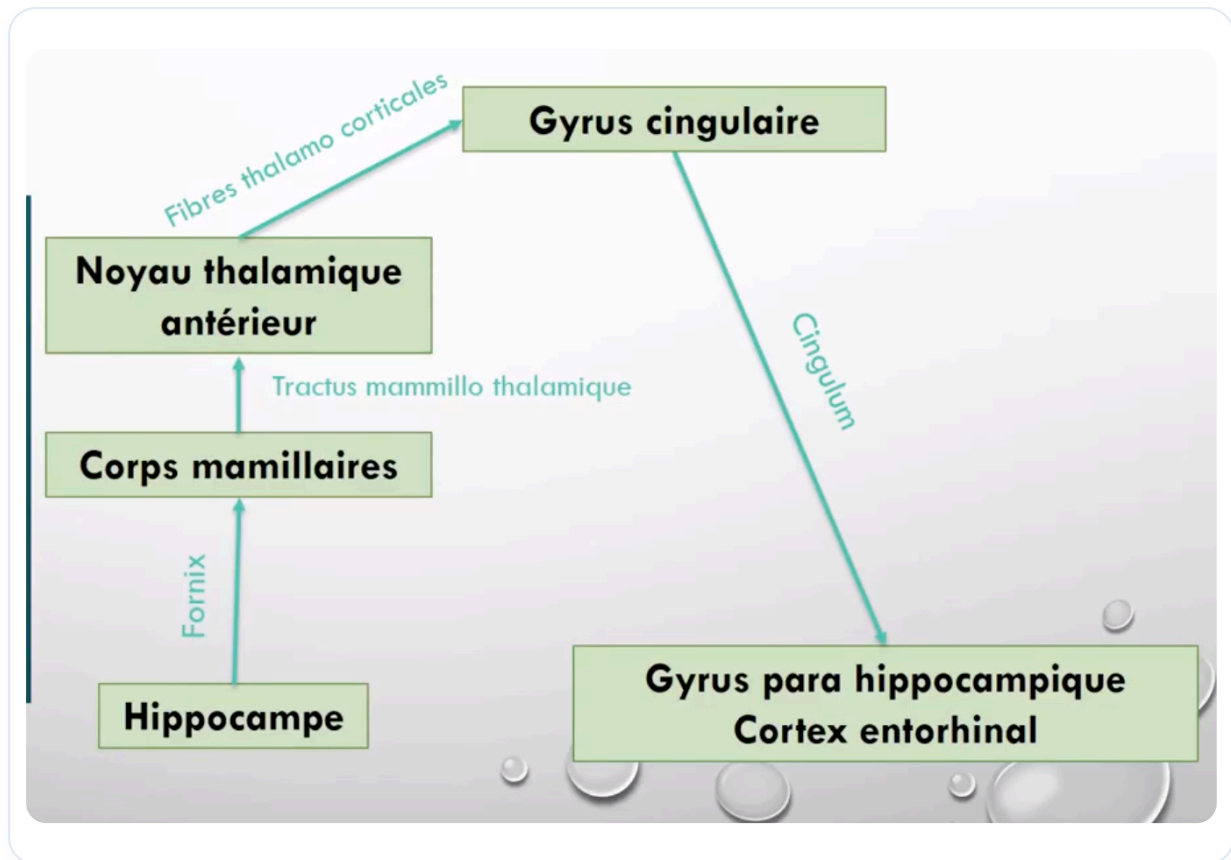


FIG. — *Role in saccades and head orientation*

INTERPRETATION OF REALITY

It is important to note that when we perceive information, we only receive a fraction of reality. Indeed, only **20%** of perceived information is real, the rest being a reinterpretation based on our experiences, emotions, and prior knowledge. This interpretation process is influenced by the connections between the visual areas and the lateral geniculate body.

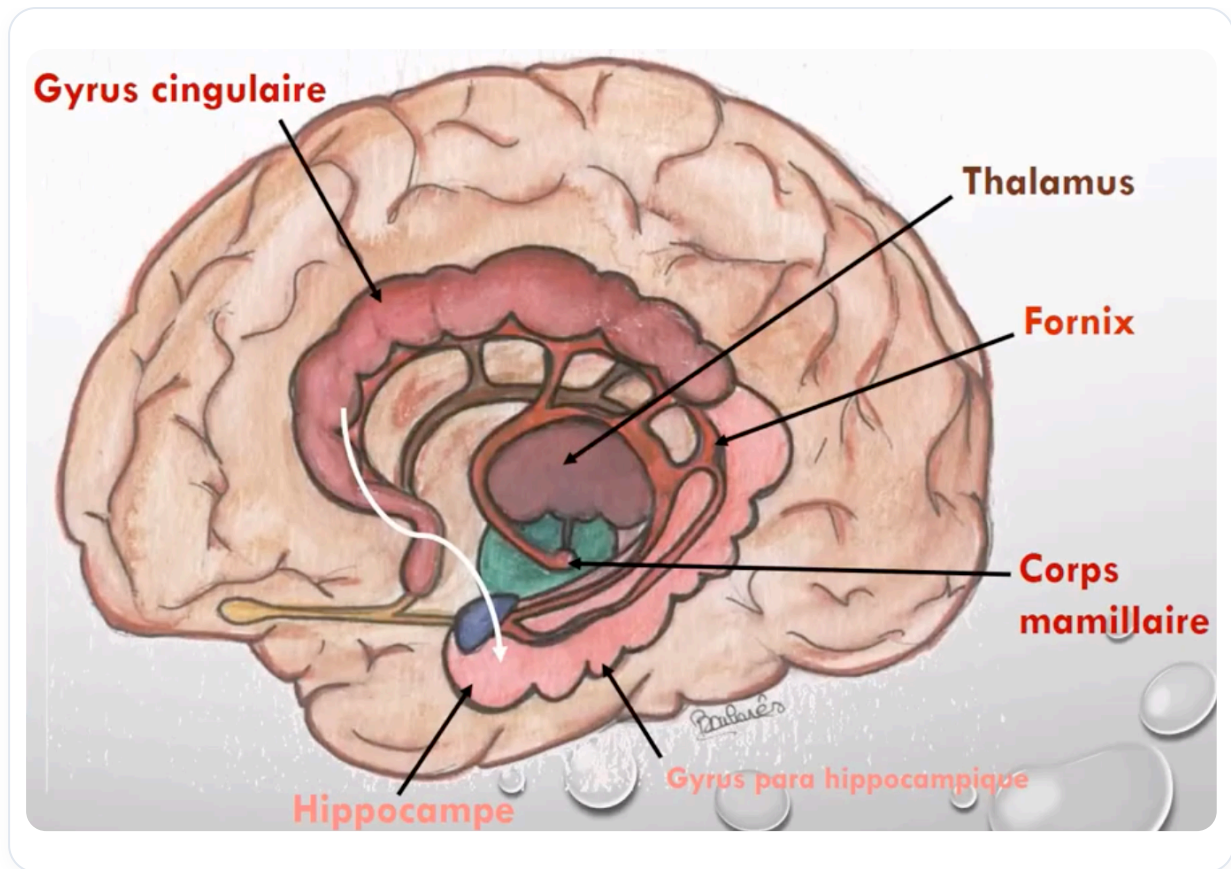


FIG. — Papez circuit and memory

CONCLUSION

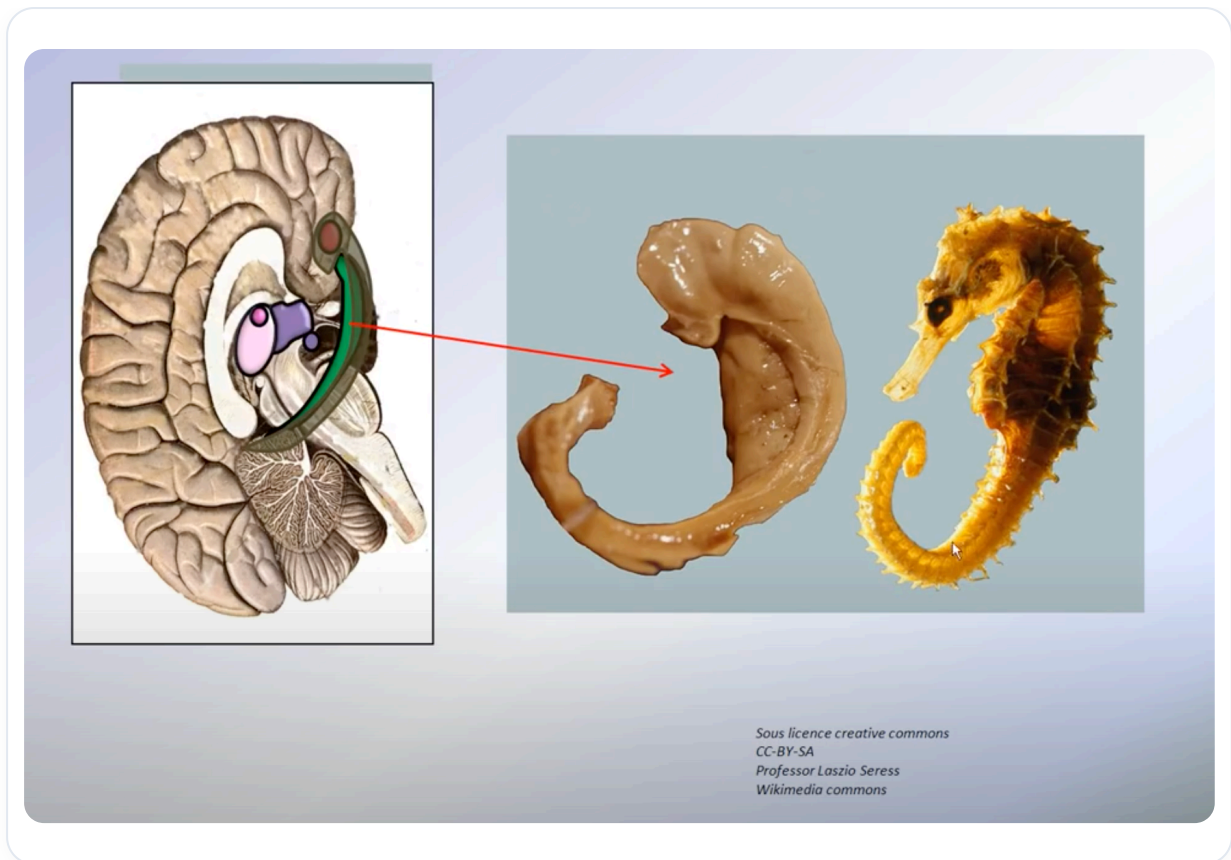


FIG. — *Papez circuit: memory consolidation*

The lateral geniculate body is a key structure in the processing of visual information, integrating data from different sensory sources and playing a crucial role in our perception of reality. Its ability to sort and interpret information is essential for our interaction with the world around us.

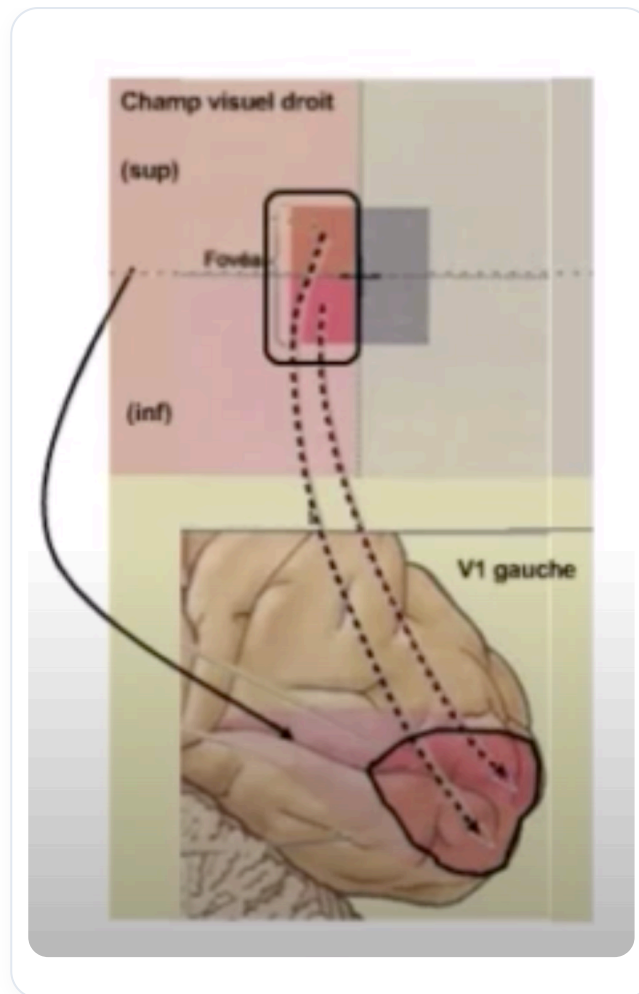


FIG. — *Reinterpretation of perceived information*

Vue interne de l'hémisphère cérébral droit

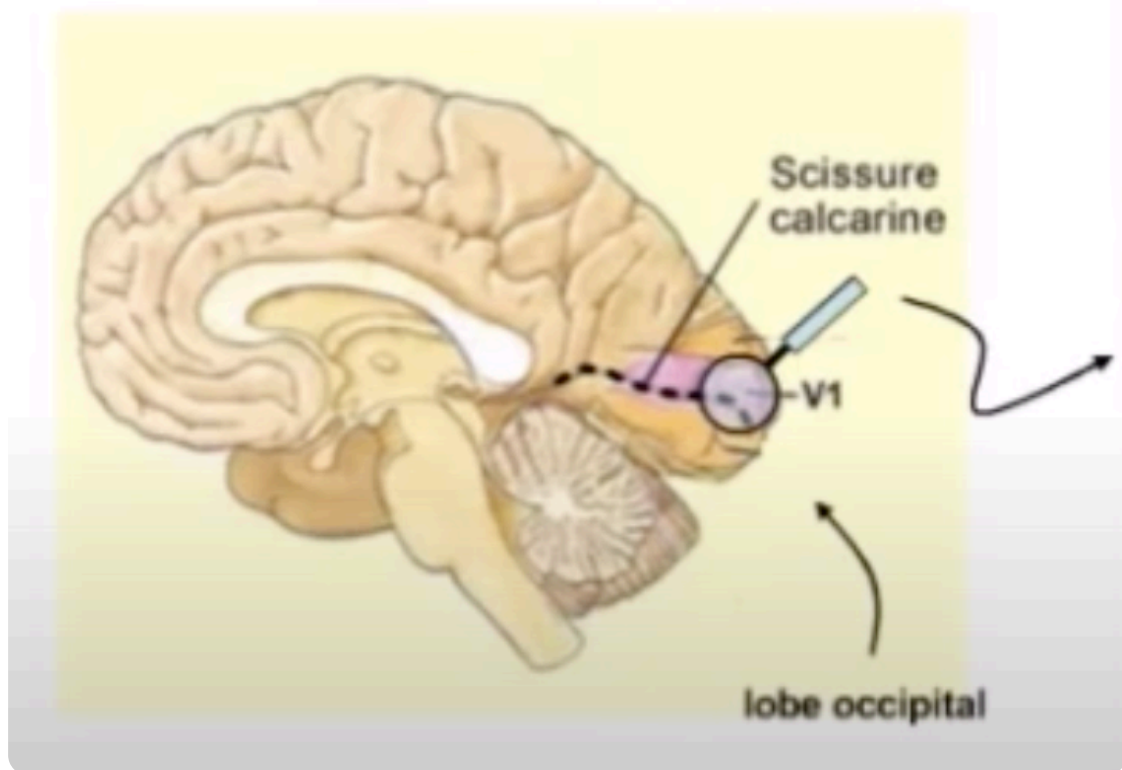


FIG. — Interpretation of reality

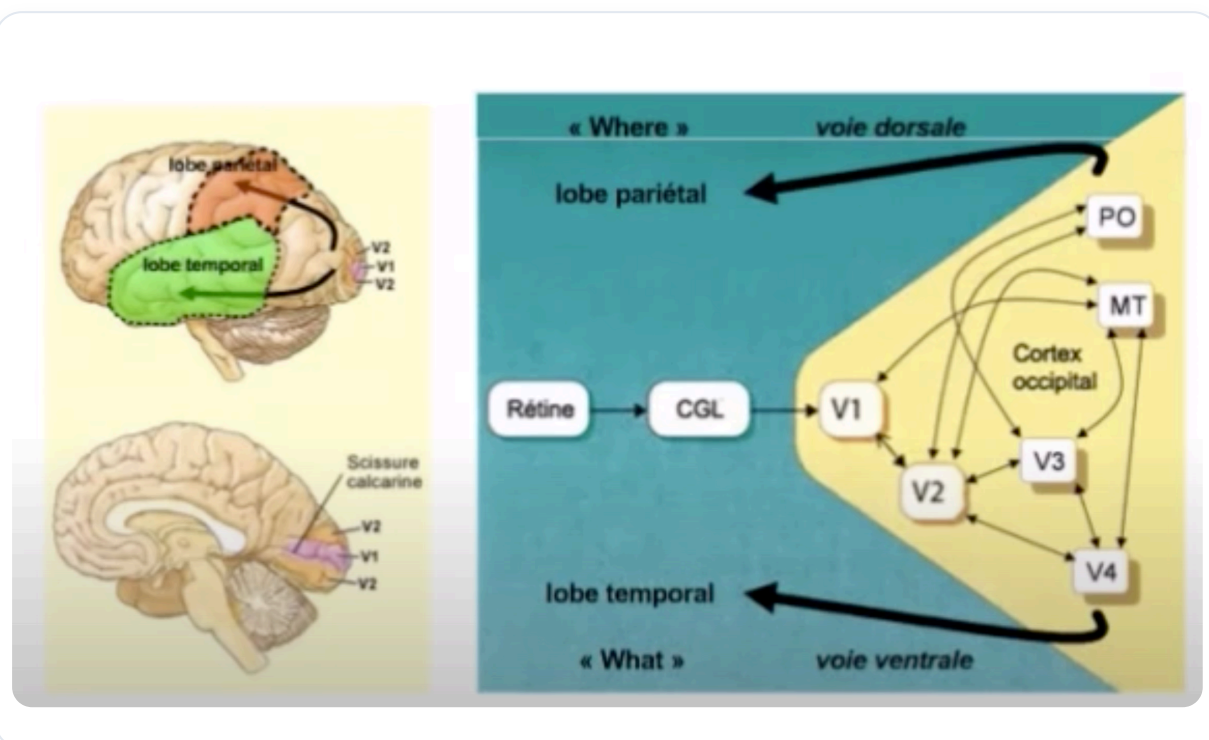


FIG. — Importance of LGN in perception

